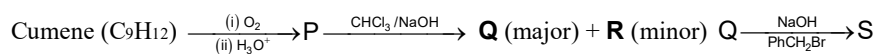
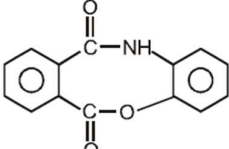


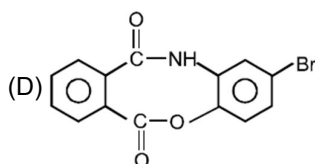
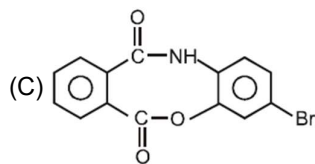
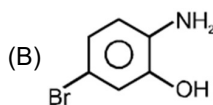
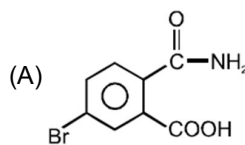
SCQ (Single Correct Type) :

1. The correct statement about the following reaction sequence is

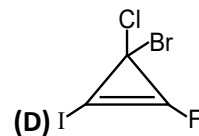
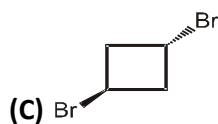
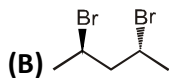
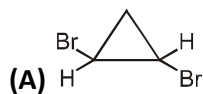


- (A) **R** is steam volatile
 (B) **Q** gives dark violet coloration with 1% aqueous FeCl_3 solution
 (C) **S** does not give yellow precipitate with 2, 4-dinitrophenylhydrazine
 (D) **S** gives dark violet coloration with 1% aqueous FeCl_3 solution

2.  'P' may be :


MCQ (Multiple Correct Type) :

3. Identify the chiral species among the following :



4. $\text{HOOC-C}\equiv\text{C-COOH} \xrightarrow[\text{CaCO}_3 \text{ Quinoline}]{\text{H}_2, \text{Pd}} \text{P (Major)} \xrightarrow{\text{Dil cold KMnO}_4} \text{Q (Major)}$

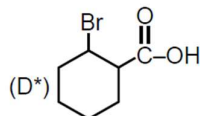
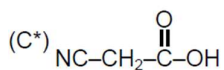
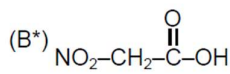
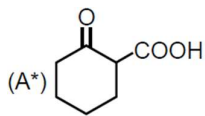
Correct statement(s) is(are) for the above reactions :

- (A) Product "P" is more acidic than its stereoisomer.
 (B) Product "Q" is optical active stereoisomer.

(C) Product "P" is malonic acid.

(D) Intramolecular H-bond takes place in conjugate base of "P".

5. Which of the following will decarboxylate easily on heating only or heating in presence of OH^-



6.

Compounds		No. of monochloro structural products after catalytic hydrogenation		Lab test for ozonolysis products [(i) O_3 , (ii) Zn , H_2O]	
(1)		(i)	5	(p)	Gives tollen's test
(2)		(ii)	6	(q)	Gives iodoform test
(3)		(iii)	4	(r)	Gives 2,4,-DNP test

Correct combination(s) is/are :

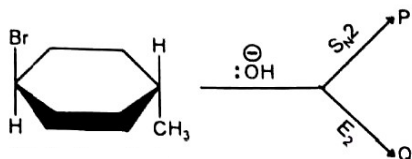
(A) (1), (i), (q,r)

(B) (2), (i), (p,r)

(C) (3) (iii) (q,r)

(D) (3), (i), (r)

7.

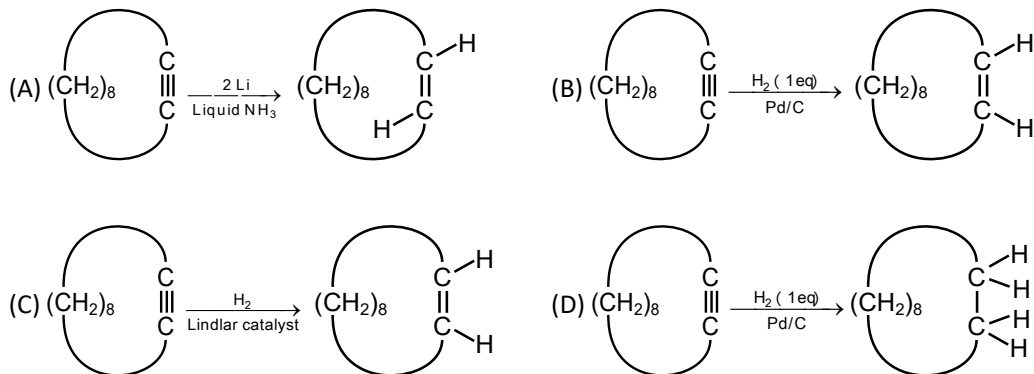


The correct statement(s) about the products of the two parallel reactions is/are:

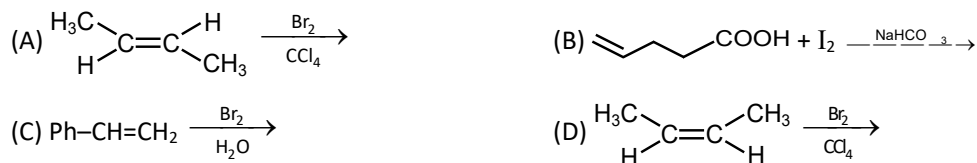
- (A) The product P is meso (optically inactive) due to internal compensation.
 (B) The product mixture Q is optically inactive.
 (C) The product P is cis isomer.
 (D) The product Q is a racemic mixture of two enantiomers.

Ans. (BCD)

8. Which of the following is written with correct product.



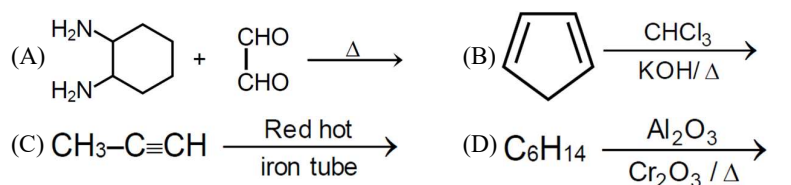
9. Which of the following reaction will form racemic mixture as major product.



10. $\text{X} (\text{C}_2\text{H}_4\text{O}_2)$ (Methyl formate) $\xrightarrow[(2) \text{H}_2\text{O}, \text{H}^+]{(1) \text{R-MgX (excess)}} \text{Y} (2^\circ \text{Alcohol})$ Possible Y is/are :



11. Which of the following may produce aromatic compound.

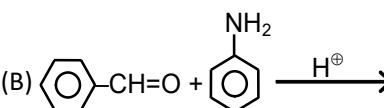


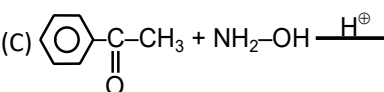
12. $\text{Ph}-\text{CH}_2\text{CH}_2\text{CH}_2\text{COCl} \xrightarrow{\text{AlCl}_3} \text{X} \xrightarrow[\text{H}^+, \Delta]{\text{Na}_2\text{Cr}_2\text{O}_7} \text{Y} \xrightarrow{\Delta} \text{Z}$



13. Observe the following reactions & choose the correct options :

(A) $\text{CH}_2=\text{CH}-\text{CH}=\text{O} \xrightarrow{\text{NaBH}_4}$ Product is unsaturated alcohol.

(B)  Product is Schiff's base

(C)  Product is two oximes

(D) $\text{R}-\text{CH}=\text{O} + 2\text{R}-\text{CH}_2-\text{OH} \xrightarrow{\text{H}^+}$ Product is Acetal

14. Which of the following sequence of reaction will synthesis biphenyl.

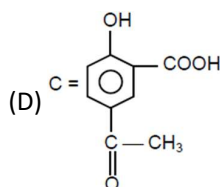
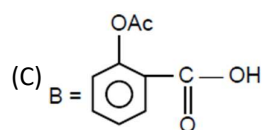
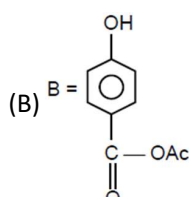
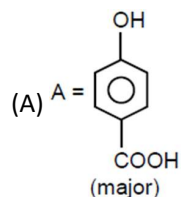
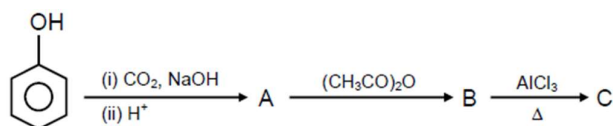
(A) $\text{C}_6\text{H}_6 \xrightarrow{\text{Br}_2/\text{Fe}} \xrightarrow[\Delta]{\text{Cu}}$

(B) $\text{C}_6\text{H}_6 \xrightarrow[\Delta]{\text{HNO}_3+\text{H}_2\text{SO}_4} \xrightarrow[\text{NaOH}]{\text{(i) Sn+HCl, (ii) HNO}_2} \xrightarrow{\text{C}_6\text{H}_6}$

(C) $\text{C}_6\text{H}_6 \xrightarrow{\text{Cl}_2/\text{AlCl}_3} \xrightarrow[\text{AlCl}_3]{\text{Ph-H}}$

(D) $\text{C}_6\text{H}_6 \xrightarrow[\text{AlCl}_3]{\text{CH}_3\text{Cl}} \xrightarrow{\text{Br}_2/h\nu} \xrightarrow[\text{AlCl}_3]{\text{Ph-H}}$

15. Choose correct options for below sequence of reaction.



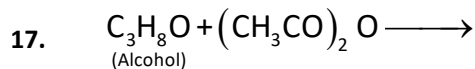
16. Which of the following is/are correctly matched?

(A) D-Glucose & L-Galactose – Epimer

(B) Fructose & Mannose – Functional isomers

(C) D-Glucose & D-Mannose – Diastereomers

(D) α -D-Glucopyranose & β -D-glucopyranose – Anomers

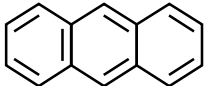
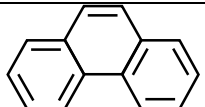
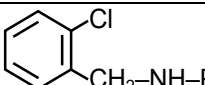
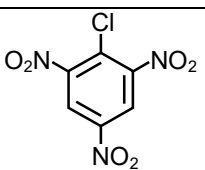


Choose the correct statement(s) about above reaction.

- (A) Products are carboxylic acid & ester.
 (B) S_N^2 th reaction.
 (C) In above reaction molecular mass of main product is increased by 42.
 (D) In above reaction three organic products are possible.

(Paragraph) :

Paragraph for Question Nos. 18 to 19

The chemistry of benzene and its derivative is dominated by one principal characteristic preservation of the aromatic π system. Electrophilic aromatic substitution is most general and important reaction of aromatic hydrocarbon. We have seen certain aromatic compounds undergo nucleophilic substitution, as well as a few other unique reactions.					
	Column-1 (Reactant)		Column-2 (Reagent)		Column-3
(I)		(i)	(1) $\text{Br}_2/\text{FeBr}_3$ (2) Base/ Δ	(P)	Addition reaction
(II)		(ii)	(1) $\text{Br}_2/\text{FeBr}_3$	(Q)	Elimination reaction
(III)		(iii)	NaNH_2 (excess) / NH_3	(R)	Overall substitution reaction
(IV)		(iv)	H_2O	(S)	Carbanion formed as one of the intermediate

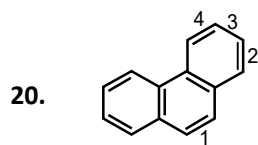
18. Both Anthracene and phenanthrene react with reagent/s to give 9-Bromo derivative as major product. Which of the following option/s has correct reagent/s and sequence of reaction mechanism/s involved in the transformation.

- (A) I (i) PQR and II (i) PQR
 (B) I (i) PQR and II (ii) PQR
 (C) I (ii) S and II (i) QR
 (D) I (ii) PQR and II (i) PQR

19. Compound (IV) reacts with a reagent forming product which is more acidic than formic acid. Which of the following option matches correctly with reagent/s and sequence of reaction mechanism involved in the transformation.

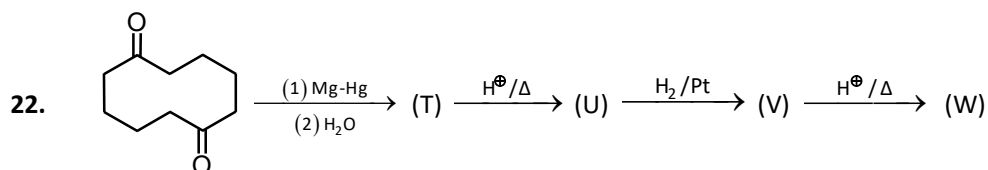
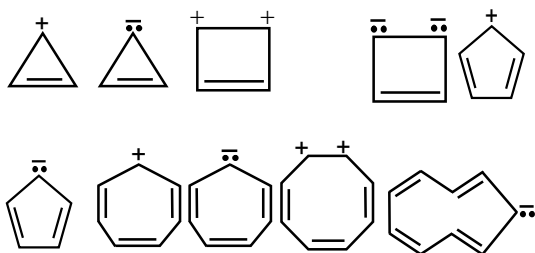
- (A) IV (iv) QPRS (B) IV (iii) PQRS (C) IV (iv) PQR (D) IV (iv) PQRS

(Numerical Value type) :



Write the number which show minimum carbon-carbon bond length in given compound :

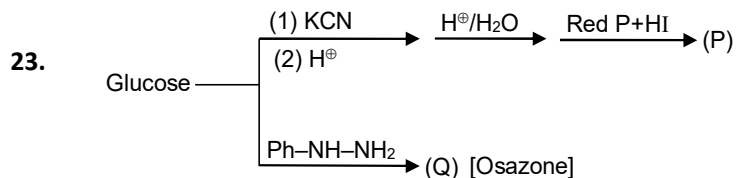
21. How many ions predicts aromatic stability according to Huckel's rule ?



(X) = Double bond equivalent (D.U.) of (W)

(Y) = Number of hyperconjugative hydrogen atom in (W)

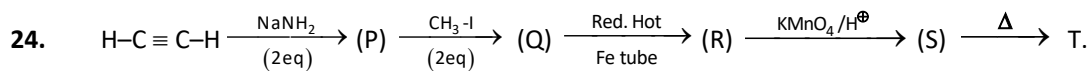
Report your answer as [X] + [Y]



[X] = Number of carbon atom in (P)

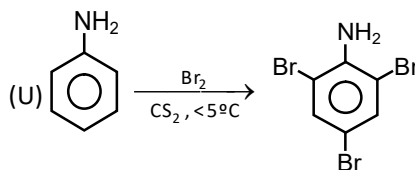
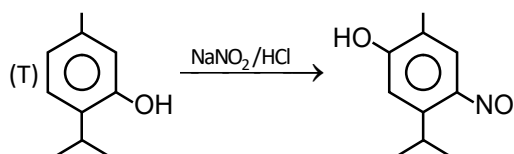
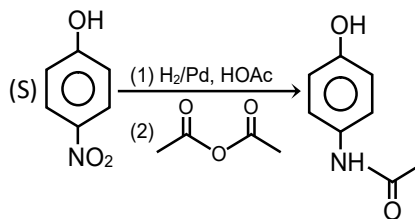
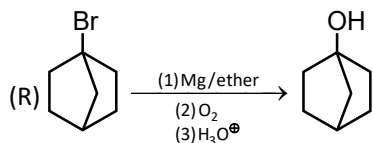
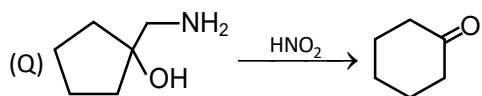
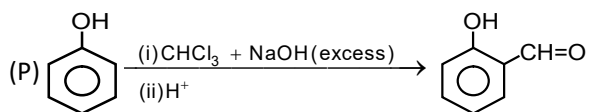
[Y] = Number of nitrogen atom in (Q)

Report your answer as [X] + [Y].



Write the number of carbon atoms present in product (T)

25. How many reactions show correct major product?



Answer key

1.	(B)	2.	(B)	3.	(ABD)	4.	(AD)	5.	(ABCD)
6.	(BD)	7.	(BCD)	8.	(ACD)	9.	(BCD)	10.	(BD)
11.	(BCD)	12.	(ACD)	13.	(ABCD)	14.	(AB)	15.	(CD)
16.	(BCD)	17.	(ABCD)	18.	(A)	19.	(D)	20.	(1)
21.	(7)	22.	(11)	23.	(11)	24.	(12)	25.	(05)

